Deep Learning on Raw CT Projection Data: A Research Proposal

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A technical research proposal — not a funding application. It makes the case for training deep neural networks on **raw CT projection (sinogram) data** rather than on reconstructed images; sets out the mathematical basis in the Radon transform and the image autocorrelation; presents preliminary results indicating the approach is feasible; and specifies what an academically-resourced investigator would need to extend it to real fan-beam / helical acquisitions and to clinically-acquired data at scale.

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1. Introduction and thesis

A modern CT scanner does not measure an image. It measures **projections** — line integrals of X-ray attenuation across a divergent (fan/cone) beam, sampled as the source rotates helically around the patient. The familiar axial image is a computed object, produced from those projections by a reconstruction algorithm that is regularized, vendor-specific, and — in practice — not invertible without loss.

The thesis of this proposal is that, for **automated diagnosis**, the projection data are the better substrate. The argument is precise and information-theoretic: because the reconstructed image is computed from the projections alone, it can carry **no more** diagnostic information than the projections already contain (the data-processing inequality, §3.5). Reconstruction is therefore, at best, information-preserving and, in reality, information-discarding: finite kernels, discretization, bit-depth quantization, apodization, and display windowing all break the equality. A network that reads the projections directly is reading the data before that loss — and before the reconstruction-introduced artifacts (beam-hardening streak, cone-beam and iterative-reconstruction textures) that never existed in the measurements.

Deep networks have already shown they can read diagnostic structure from projections: several groups have learned the sinogram→image reconstruction map directly (AUTOMAP; Li, Chen et al.; DEAR — §2). Those networks learn to make a better image. This proposal reuses the same projection-domain modeling capability for a **different endpoint — the diagnosis itself** — and never forms the image. The exemplar task throughout is detection of acute **intracranial hemorrhage (ICH)**, a time-critical finding for which the earliest possible flag has clinical value.

The proposal is deliberately honest about what present evidence can and cannot show. Real CT is fan-beam and helical; the public data used for the feasibility experiments provide reconstructed images only, so the experimental sinograms are **synthetic parallel-beam re-projections**. That regime rigorously tests **representation and architecture** — can a network detect ICH from a projection encoding, and how efficiently — but it cannot demonstrate the raw-data fidelity and artifact advantages, which are properties of true measured projections. Separating these two regimes is the central discipline of the argument, and defines the work that remains (§6).

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2. Historical background

The mathematics predates the machine by half a century. **Johann Radon** (1917) proved that a function on the plane is determined by its integrals over all lines, and gave the inversion — the theoretical core of tomography. **Ronald Bracewell** (1956) independently used projection–reconstruction ("strip integration") to map the radio sky, and articulated the **Fourier-slice relationship**. **Dennis Gabor** (1946) introduced the jointly space/frequency-localized "elementary signals" now called Gabor atoms in his Theory of Communication; Gabor also invented **holography** (Nobel Prize, 1971).

The clinical machine followed from **Allan Cormack** (1963–64), who worked out the radiological line-integral inversion, and **Godfrey Hounsfield** (1971–73), who built the first scanner at EMI; the two shared the 1979 Nobel Prize in Physiology or Medicine. The canonical mathematical treatment — parallel- and fan-beam filtered back-projection, algebraic reconstruction — is Kak & Slaney (1988). **Helical (spiral) CT** (Kalender et al., 1990) made the source trace a helix relative to the patient; exact inversion for the helical cone beam came only later (**Katsevich**, 2002). The autocorrelation–power-spectrum identity underlying the feature argument of §3.3 is the **Wiener–Khinchin** theorem (Wiener 1930; Khinchin 1934).

The deep-learning era reopened the inverse problem: **AUTOMAP** (Zhu et al., Nature 2018) learned image formation as a manifold transform; **Li, Chen et al.** (IEEE TMI 2019) learned the sinogram→image map directly on CT; **Xie, Shan & Wang** (DEAR, 2019) pursued few-view reconstruction. Each learns to make a better image. The present proposal reuses the same projection-domain modeling capability for a different endpoint — the diagnosis — and never forms the image.

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3. Mathematical framework

3.1 The real acquisition geometry: fan-beam and helical

A clinical scanner is **not** a parallel-beam Radon device. A point source illuminates a detector arc, so the measured rays are **divergent (fan-beam)**: a projection is $g(\beta, \gamma)$, with β the source angle and γ the fan angle of a ray. Fan data relate to parallel data by a rebinning,

$$\theta = \beta + \gamma, \; s = D \sin \gamma,$$

with D the source-to-isocenter distance; fan-beam FBP additionally carries a distance weighting. Acquisition is **helical**: the table translates while the gantry rotates, so the source describes a helix relative to the patient,

$$z(\beta) = (h / 2\pi)\, \beta \text{ (pitch } h \text{ per rotation).}$$

Reconstructing an axial plane then requires longitudinal interpolation (single-slice helical), cone-beam methods (FDK), or exact helical inversion (Katsevich 2002), and in practice increasingly **model-based iterative** or **deep-learning** reconstruction. The operational point: **raw data are divergent-beam, helically sampled, and reconstruction is a regularized, vendor-specific, non-injective-in-practice inverse.** That non-injectivity is exactly where kernel choice, apodization, and iterative priors inject the image-domain variability that projection-domain detection avoids.

3.2 The parallel-beam Radon transform (an idealization we use to reason)

The parallel-beam transform is the tractable model that exhibits projection structure. For attenuation field f,

$$p_\theta(s) = \int f(s \cos\theta - t \sin\theta, s \sin\theta + t \cos\theta)\, dt.$$

A point mass at polar position (r_0, φ_0) maps to $s = r_0 \cos(\theta - \varphi_0)$ — a sinusoid, hence sinogram. The **Fourier-slice theorem** connects it to the image spectrum F:

$$(\text{1-D FT of } p_\theta)(\omega) = F(\omega \cos\theta, \omega \sin\theta),$$

so projections tile the 2-D spectrum radially and f is recoverable with full angular coverage. Inversion is filtered back-projection,

$$f(x,y) = \int_0^\pi (p_\theta * H)(x \cos\theta + y \sin\theta)\, d\theta, \quad H(\omega) = |\omega|,$$

whose ramp filter $|\omega|$ amplifies high-frequency noise — the ill-conditioning that forces the regularization choices of §3.1. This parallel-beam model is an **illustrative, computable stand-in** for the projection structure; it is not the acquisition physics, and the synthetic data of §5 inherit its idealization.

3.3 The sinogram encodes the image autocorrelation

The feature the detector must read is second-order. Define the image autocorrelation

$$R_f(\tau) = \int f(x)\, f(x + \tau)\, dx, \quad (\text{2-D FT of } R_f) = |F|^2 \text{ (Wiener–Khinchin).}$$

The 1-D autocorrelation of one projection,

$$r_\theta(u) = \int p_\theta(s)\, p_\theta(s + u)\, ds, \quad (\text{1-D FT of } r_\theta)(\omega) = |F(\omega \cos\theta, \omega \sin\theta)|^2,$$

is, by Fourier-slice applied to $|\cdot|^2$, a **radial slice of the 2-D power spectrum**. Sweeping θ fills $|F|^2$, whose inverse transform is R_f. In words: **the projection autocorrelations, across angle, reconstruct the image autocorrelation** — each projection is an angular sample of the object's correlation/texture structure. A hemorrhage — a focal, hyperattenuating, texturally distinct region — perturbs R_f in an orientation-dependent way that is present, angularly encoded, in the sinogram.

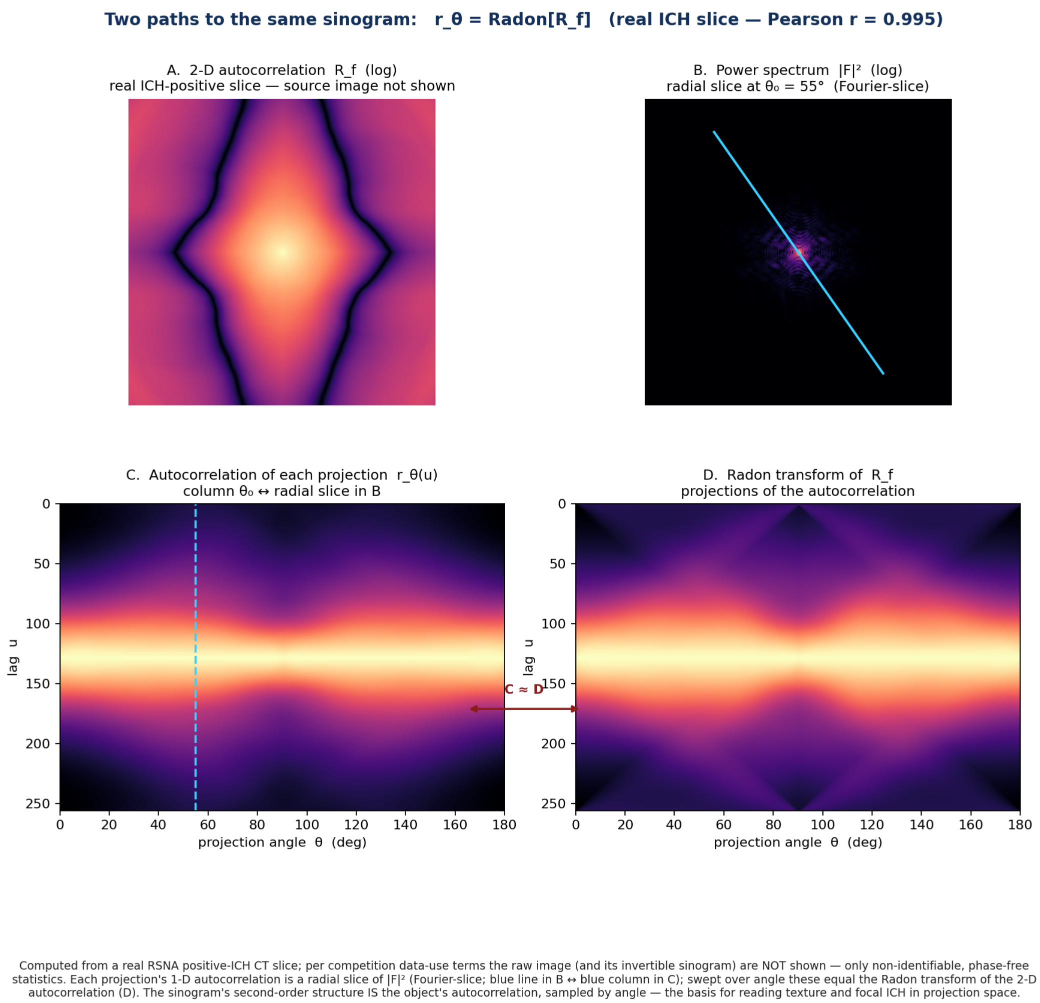


Figure 1. The autocorrelation identity, computed on a **real RSNA positive-ICH slice** (the raw image and its invertible sinogram are not shown — only non-identifiable, phase-free statistics, per competition data-use terms). **(A)** 2-D autocorrelation R_f (log). **(B)** Power spectrum $|F|^2$ with a radial slice at $\theta_0 = 55°$ (Fourier-slice). **(C)** The 1-D autocorrelation of each projection, $r_\theta(u)$, stacked over angle; the marked column corresponds to the radial slice in B. **(D)** The Radon transform of the 2-D autocorrelation R_f. Panels C and D are equal (Pearson $r = 0.995$), demonstrating $r_\theta = \text{Radon}[R_f]$: the sinogram's second-order structure is the object's autocorrelation, sampled by angle.

3.4 Gabor wavelets: the natural feature basis

A Gabor atom is a Gaussian-windowed oriented sinusoid,

$$\psi(x; f_0, \theta, \sigma, \varphi) = \exp(-|x|^2 / 2\sigma^2) \cdot \cos(2\pi f_0 (x \cdot n_\theta) + \varphi),$$

jointly localized in space and frequency (attaining the uncertainty bound; Gabor 1946) and selective in orientation θ and scale $\propto 1/f_0$. Two facts make it the right basis here. First, an oriented image structure maps to a localized (θ, s) signature in the sinogram, so a Gabor bank over the sinogram reads projected orientation and scale directly — Gabor energy is a **localized power spectrum**, complementing the global autocorrelation of §3.3. Second, the first convolutional layers of trained CNNs are empirically Gabor-like; a CNN on the sinogram is, in effect, learning a **data-adaptive Gabor/autocorrelation front end**. A projection-domain detector therefore formalizes what its early layers would learn regardless

— usable both as initialization (a fixed Gabor front end) and as an interpretability handle.

3.5 Information ordering — and an honest caveat

For a real scan the causal chain is

$$Y \text{ (diagnosis)} \to f \text{ (attenuation field)} \to p\_raw \text{ (fan-beam helical data)} \to x \text{ (reconstructed image)},$$

with x computed from p_raw alone, so (Y, p_raw, x) form a Markov chain. The **data-processing inequality** gives

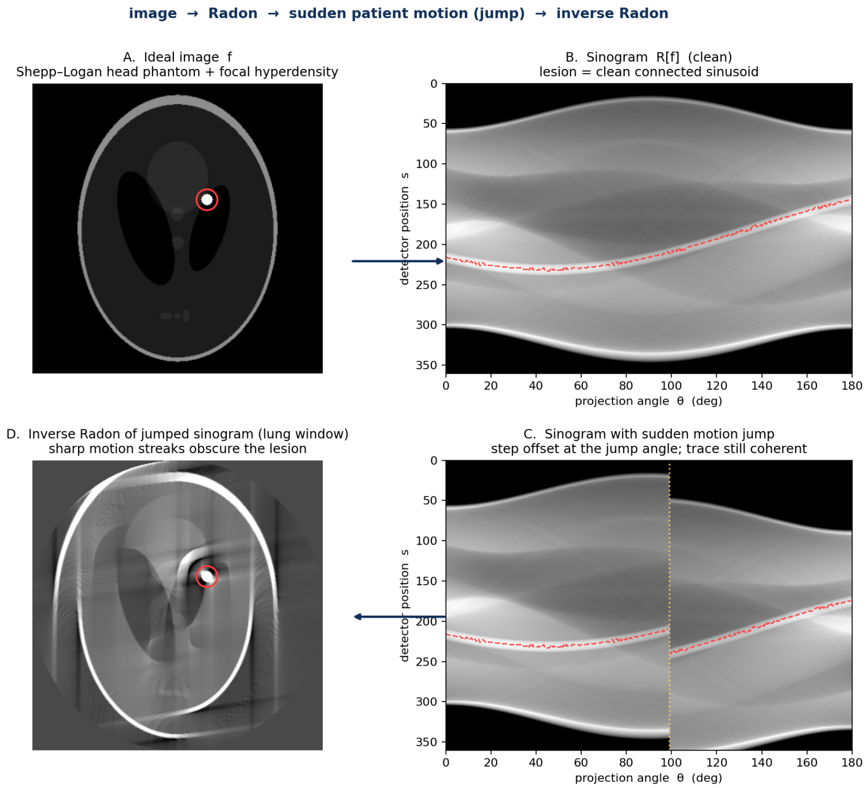
$$I(Y; x) \leq I(Y; p\_raw).$$

Reconstruction cannot increase the diagnostic information the data carry; equality requires reconstruction to be a sufficient statistic for Y, which finite kernels, discretization, quantization, apodization, and display windowing all break. **This is the rigorous form of the fidelity claim.**

But public archives provide x only. Synthetic sinograms are **produced by forward projection**, p_syn = R[x], so the chain is Y → x → p_syn and

$$I(Y; p\_syn) \leq I(Y; x) \leq I(Y; p\_raw).$$

The synthetic sinogram is a re-encoding of the already-reconstructed image — it cannot exceed it — and it is parallel-beam, not fan-beam helical. **Consequently synthetic-sinogram experiments can rigorously establish representation and architecture claims and simulate geometry/dose robustness, but they cannot by themselves demonstrate the raw-domain fidelity or artifact advantages.** Those require true p_raw (§6).



A sudden mid-scan patient jump is a single step discontinuity in the sinogram (C, yellow line): the lesion track offsets abruptly but stays a clean, connected sinusoid. Back-projecting that discontinuity, however, throws sharp radiating streaks across the reconstructed image (D, lung window) that bury the same lesion. Detecting the lesion in sinogram space can thus be more robust to motion than detecting it in the reconstructed image.

Figure 2. The image↔projection loop and the motion argument. **(A)** An ideal image f (Shepp–Logan head phantom with a focal hyperdensity, red circle). **(B)** Its sinogram R[f]; the lesion is a clean, connected sinusoid (red track). **(C)** A sudden mid-scan patient motion, modeled as a step offset of the sinogram past the jump angle (yellow line) — the lesion trace offsets abruptly but each segment remains a coherent sinusoid. **(D)** Inverse Radon of the motion-corrupted sinogram, shown in a wide, low-centered lung window: the step discontinuity back-projects into sharp radiating streaks that bury the lesion. The abnormality's signature is better preserved in projection space (B, C) than in the reconstructed image (D), which motivates detection before reconstruction. All panels are real Radon/inverse-Radon transforms; the lesion track is the argmax ridge of the lesion-only sinogram under the same step offset.

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4. Advantages of the projection-domain approach

Each advantage below is, at root, a property of true raw projections. The synthetic experiments of §5 can probe some of them by simulation but cannot fully demonstrate them.

1. **Reconstruction artifacts are avoided.** Beam-hardening, streak, cone-beam and iterative-reconstruction textures are introduced by reconstruction; a sinogram reader never sees them. (Approachable by simulation; confirmable on raw data.)
2. **Higher intrinsic fidelity.** Raw projections carry greater bit depth and finer detector sampling than the windowed, resampled image retains — formalized by the data-processing inequality of §3.5. (A raw-data property; not demonstrable on synthetic sinograms.)

3. **Freedom from cross-algorithm inaccuracy.** An image-domain model inherits kernel/slice/vendor dependence — a leading cause of generalization failure across reconstructions. A projection-domain detector is instead calibrated to one specific scanner's raw output; the reconstruction-algorithm variable is removed, not averaged over. (Partly testable by simulated-geometry robustness; confirmable on raw data.)
4. **Speed of notification.** Detection precedes or runs concurrently with reconstruction, enabling the earliest physically possible flag for a finding in which minutes change neurological outcome. (Latency measurable now; end-to-end only on-scanner.)

A concrete clinical case — the posterior fossa. Reconstruction-introduced artifacts are not uniform across the brain: beam-hardening streaks (trans-petrous / interpetrous) and dental artifact fall hardest on the posterior fossa and brainstem — precisely where small hemorrhages are most easily missed on CT, and where MRI is well established as superior. Because a projection-domain reader never sees a reconstruction-introduced artifact, the approach's clinical value may be largest exactly where CT is weakest. This yields a falsifiable prediction, testable on a suitably labeled clinical cohort: a projection-domain detector should show a disproportionate sensitivity advantage for **infratentorial** (brainstem / posterior-fossa) hemorrhage relative to supratentorial, versus an image-domain detector.

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5. Feasibility: preliminary results

All results below are on **synthetic parallel-beam sinograms** forward-projected from the RSNA 2019 Intracranial Hemorrhage dataset (>870,000 expertly labeled head-CT slices; six labels per slice — epidural, intraparenchymal, intraventricular, subarachnoid, subdural, and any). Each slice is encoded on a calibrated linear-Hounsfield scale (value = HU + 1024) and forked into a reconstructed branch (256×256, 12-bit) and a synthetic 256-angle sinogram (256×256, float32) from one shared source. Stratified cohorts of 50,000 and 200,000 slices were built with zero processing failures, preserving the full-dataset hemorrhage prevalence (14.5% positive) and all 32 observed label-combinations proportionally. A **twin protocol** underlies every comparison: an identical architecture and initialization is trained on each domain, differing only by input, so any difference is attributable to the domain. Work ran on a single-GPU workstation.

Detection is feasible and matches the image domain. A projection-domain network detects ICH at best validation **macro-AUC 0.946** across the six classes (SE-ResNeXt50, 200k), statistically on par with the matched image-domain twin. This answers the primary representation question affirmatively: a network can read ICH from a projection encoding, at image-domain-competitive accuracy.

Performance still climbs with data. A whitened data-size sweep shows both domains improving monotonically with N, the projection-domain model matching or marginally edging the image-domain model throughout (image-domain vs projection-domain macro-AUC):

- **N = 50,000:** 0.807 vs 0.824.
- **N = 100,000:** 0.832 vs 0.835.
- **N ≈ 180,000:** 0.847 vs 0.851.

The curve has not plateaued at ~180k — directly relevant to the data-scale argument of §6.

The learned representation carries a domain signature. With small local kernels on raw input, first-layer filters are indistinguishable between domains (both become generic oriented high-pass filters, AUC ≈ 0.81). But with large (31×31) stems and the input's average power spectrum **whitened** — removing the low-order statistics a network fits first — a clear domain signature emerges: sinogram-native kernels are **anisotropic, detector-axis-oriented, and lower-frequency**; image-domain kernels are isotropic and higher-frequency. Notably, the literal filtered-back-projection ramp (§3.2) does **not** appear — the network learns Radon-geometry-aligned structure directly, not textbook back-projection. This motivated a **sinogram-native anisotropic architecture** whose stem geometry is matched to the projection trace; an architecture search over such designs found a configuration with a small but **statistically robust** gain over the baseline (+0.005 macro-AUC; paired t = 3.4 over six seeds, Cohen's d = 1.4).

Angular robustness is a real but conditional advantage. Under simulated angular under-sampling (a removed wedge of projection angles, applied natively in each domain), a controlled factorial over augmentation × spectral-whitening × domain × model-capacity establishes that the projection domain's graceful degradation is **not** an unconditional intrinsic property. Much of it is a trained property — reproducible in the image domain once that branch is shown the matched physical corruption — and the residual projection-domain advantage is conditional: clearly present at moderate angle loss and high model capacity (retention 0.81 vs 0.57 for the image-domain model at a 30% wedge), but absent at severe angle loss, where both domains collapse without augmentation. This result was obtained as a self-imposed negative control — the apparent advantage was first traced to, and separated from, a training-augmentation confound before the conditional advantage was affirmed.

Summary. The representation-and-architecture question is answered: ICH is detectable from a projection-domain encoding at image-domain-competitive accuracy, the network's learned features are interpretable and geometry-aligned, and a projection-native architecture measurably helps. What synthetic parallel-beam data cannot show — the fidelity and artifact advantages of §4 — is the object of §6.

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6. Directions for further investigation

The feasibility evidence is a floor, not a ceiling. Realizing the argument requires three extensions, each needing resources beyond an individual investigator's reach — which is the honest reason this is offered as an open research proposal that an academically-credentialed group could take up.

6.1 From synthetic parallel-beam to real fan-beam / helical geometry. The experiments of §5 use a parallel-beam idealization. The immediate technical work is a **fan-beam helical acquisition-and-reconstruction pipeline** — helical→fan rebinning (§3.1) and a differentiable reconstruction layer — feeding the same detector head, so that the representation results can be reproduced under real geometry.

6.2 Reconstruction-fidelity validation against measured projections. The public **Mayo Clinic / AAPM** LDCT-and-Projection-data collection (TCIA) provides real measured **DICOM-CT-PD** helical projections with paired reconstructions for ~99 non-contrast head, 100 chest, and 100 abdomen exams, on Siemens and GE scanners. This is the natural instrument to **check reconstruction and forward-projection fidelity**: forward-project the collection's reconstructed images through the documented scanner geometry and compare against the actual measured projections, quantifying how faithfully a simulator reproduces real raw data. It requires no manufacturer agreement — only a data-use registration. It is, however, a fidelity-validation cohort, not a training cohort: ~99 head exams contain far too few hemorrhages to train a detector.

6.3 The data-scale problem, and why academic/institutional access is essential. Training a projection-domain ICH detector to clinical performance is a large-data problem, and the §5 sweep — still rising at ~180k — indicates the requirement is not small. A production-grade detector across six subtypes (several individually rare) plausibly needs on the order of **200,000 studies**. At the ~14% ICH prevalence of the reference cohort, 200,000 studies yield roughly 28,000 positive studies — and far fewer for the rarest subtypes; in unselected consecutive clinical practice the positive yield is typically lower than this curated figure, which only raises the number of studies required.

The binding constraint is subtler than volume: **raw projection data are not retained.** Scanners discard the projections after reconstruction, and PACS stores only images — so a large raw-data cohort cannot be mined retrospectively; it must be **prospectively retained at acquisition**, which requires both institutional data-governance and manufacturer cooperation. Under prospective accrual, a single large academic medical center performing on the order of ~50,000 head CTs per year would need roughly **four years** to reach 200,000 studies; a consortium of several large academic facilities could assemble a comparable cohort in about **one year**, which is the realistic route to an initial feasibility dataset. An investigator with academic credentials and institutional standing — able to secure IRB approval, multi-site data-governance, and a manufacturer raw-data agreement — is therefore a prerequisite, not a convenience.

6.4 Further technical questions worth pursuing.

- **Where the global trace assembles.** A first-layer kernel cannot see a full sinusoidal lesion trace; an effective-receptive-field / deeper-layer analysis should locate where the network integrates the trace across angle.
- **A reconstruction / self-supervised objective.** Classification never produced the literal FBP ramp; a reconstruction objective should, and would test whether the ramp is latent but unused.
- **Fixed-Gabor vs learned front end**, and an explicit autocorrelation feature (§3.3–3.4), as both initialization and interpretability probes.
- **Fair image-domain comparison under realistic artifacts.** The image-domain twin here was trained on clean synthetic reconstructions; a fair contest should expose it to realistic streak/metal/motion artifacts, which real clinical images contain and which may narrow — or, per §4, widen — the projection-domain advantage, especially infratentorially.
- **Back-projected saliency maps** to render projection-domain decisions in anatomic space for clinical interpretability.
- **The infratentorial-sensitivity test** (§4), requiring anatomic localization of bleeds beyond the subtype labels of the reference dataset.

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7. Conclusion

Reconstruction is a lossy, vendor-specific, artifact-introducing inverse standing between the measurement and the diagnosis. The information-theoretic argument, the autocorrelation/Gabor feature basis, and — now — preliminary results showing image-domain-competitive ICH detection from synthetic sinograms together make a coherent case that **diagnosis is better posed on the raw projection data than on the reconstructed image**. The representation question is answered; the remaining advantages are properties of real measured data, reachable only with fan-beam/helical raw acquisitions and clinically-acquired cohorts at academic scale. The purpose of this document is to put that case, and that remaining program, on the record clearly enough for a suitably resourced investigator to take it up.

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Classical citations are standard references from the canonical literature; page-level details should be verified against the originals. Deep-learning and dataset citations were confirmed against publisher / PubMed / archive records.

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